

Financial GraphRAG

C-09



Literature Review

Paper	Methodology	Dataset Used	Key Results	Limitations
GraphRAG: From Local to Global (Microsoft, 2024)	Builds an LLM-generated Knowledge Graph from documents → performs entity & relation extraction → applies Leiden community detection → generates	Large private document corpora (~1M token scale); evaluated on benchmark datasets for query-focused summarization rather than finance-specific data.	Outperformed vanilla RAG in answer comprehensiveness and answer diversity for global questions over large corpora. Also achieved competitive performance with lower token cost than global map-reduce summarization approaches. (GraphRAG)	Not designed for structured financial data; optimized for global summarization, not numerical reasoning or financial QA. Graph construction is computationally
Think-on-Graph (ToG, 2023)	Treats the LLM as a graph-traversal agent. Instead of retrieving a fixed subgraph, the LLM iteratively explores entities and relations, performs beam search, prunes	Evaluated on several knowledge graph QA benchmarks (general-domain KGs such as Wikidata/Freebase-style datasets), not financial datasets.	Consistently improved multi-hop reasoning accuracy over conventional KGQA and RAG baselines while providing interpretable reasoning paths. (GraphRAG)	Evaluated only on general-domain knowledge graphs. Does not address financial reports, temporal filings, or numerical reasoning. Beam-search traversal can become
HybridRAG (2024)	Combines Vector Retrieval + Knowledge Graph Retrieval. Retrieves semantic context from a vector database while simultaneously retrieving	Financial earnings call transcripts (Nifty-50 companies) with ground-truth Q&A pairs extracted from transcripts.	Demonstrated better retrieval quality and answer generation than using VectorRAG alone or GraphRAG alone, showing that hybrid retrieval is more effective for financial QA. (GraphRAG)	Uses a fixed retrieval pipeline (no adaptive routing or agent planning). Limited to earnings calls rather than comprehensive SEC filings.
FinReflectKG (2025)	Proposes an agentic KG construction pipeline: intelligent document parsing → table-aware chunking → schema-guided extraction → reflection agent	SEC 10-K filings from S&P 100/500 companies (2014–2024). Final graph contains ~17.5 million triples with provenance links to source chunks.	Reflection-agent extraction achieved 64.8% rule compliance, a 53.2% relative improvement over single-pass extraction. Produces a large, auditable financial KG. (arXiv)	Focuses on building the KG, not on question answering. Quality still depends on LLM extraction and schema design.
FinReflectKG-MultiHop (2025)	Builds a multi-hop financial QA benchmark by mining 2–3 hop subgraphs from FinReflectKG. Creates analyst-style questions with KG-grounded evidence and	Built on FinReflectKG, covering S&P 100 companies (2022–2024). Released benchmark subset contains 555 QA pairs spanning intra-document, inter-year, and	KG-guided retrieval improved correctness by ~24% while reducing token usage by ~84.5% compared with page-window retrieval, across both reasoning and non-reasoning LLMs. (arXiv)	Benchmark only—it does not propose a QA system. Evaluation focuses on retrieval settings rather than adaptive agents or routing strategies.

Objectives

- Develop an agentic retrieval system over the FinReflectKG knowledge graph, combining graph traversal, hybrid passage search, and symbolic computation to answer multi-hop financial questions from SEC 10-K filings.
- Design a query-adaptive retrieval controller that selects lightweight or comprehensive retrieval based on estimated question complexity, rather than applying one uniform strategy to every query.
- Evaluate the system across staged retrieval configurations to quantify how much each component (retrieval, graph enrichment, adaptive routing) contributes to answer accuracy and reasoning-type performance.
- Identify and mitigate discrepancies between retrieval quality and reasoning accuracy by improving how retrieved evidence is structured and presented to the model.
- Measure system efficiency and answer traceability — latency/compute savings from adaptive routing, and how well generated answers can be traced back to retrieved evidence.

Cross-domain technique adoption

- **Claim-level provenance verification:-** Decompose generated answers into individual atomic claims and verify each one separately against retrieved KG triples, rather than scoring the whole answer at once, catching partial errors that answer-level checks miss. (from MedRAGChecker)
- **Confidence-scored routing signal:-** Use confidence scores on extracted graph entities/relations as one input to the routing decision — low-confidence or sparse evidence pushes a query toward full graph traversal instead of lightweight retrieval. (from AMG-RAG)
- **Personalized PageRank as complexity estimator:-** Score how well-connected a query's relevant facts are within the graph, and use that connectivity signal to decide whether the query is simple (tight neighborhood) or complex (sparse/spread-out evidence). (from HippoRAG)

Progress:-

```
● PS A:\S7\Phase Project> python check_stats.py
Total Nodes: 15663
Total Relationships: 51927

Node Counts by Entity Type:
  FIN_METRIC: 6088
  PRODUCT: 1410
  RISK_FACTOR: 1098
  ACCOUNTING_POLICY: 1080
  FIN_INST: 999
  CONCEPT: 726
  REGULATORY_REQUIREMENT: 616
  COMMENTARY: 477
  COMP: 461
  MACRO_CONDITION: 378
  SEGMENT: 318
  EVENT: 297
  LITIGATION: 274
  LOGISTICS: 236
  PERSON: 173
  GPE: 158
```

Ln 19, Col 1 Spaces: 4 UTF-8 LF { } Python 3.12.3 64-bit Select Python Interpreter Pyrefly (Basic) Antigravity - Settings



🏠 Get started

</> Developer hub

Data services

📁 Instances

🔗 Graph Analytics

🔗 GraphQL Data APIs

📦 Agents New

Studio

📁 Import

📄 Query

🔗 Bloom

📊 Dashboards

📄 Document Intelligence

📈 Operations >

📁 Project >

📖 Learning

Instances

Aura (1)

Self-managed (0)

Virtual Graphs

Preview

🔍 Search

📄 88

Create instance

GraphRAG

● RUNNING

ID: 1b9a653f 📄

Connect ▾

⋮

Type: AuraDB Free Nodes: 15,663 (8%) Relationships: 51,927 (13%)

Upgrade

▾ Metrics



2-HOP & 3-HOP PATTERNS

=== Frequent 2-hop patterns in your graph ===

```
(PERSON) -[has_stake_in]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=44799
(PERSON) -[discloses]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=39113
(PERSON) -[announces]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=33427
(ORG_REG) -[regulates]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=31060
(PERSON) -[member_of]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=19830
(PERSON) -[works_for]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=16127
(FIN_METRIC) -[impacted_by]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=11898
(FIN_MARKET) -[negatively_impacts]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=11898
(PERSON) -[join]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=11230
(COMP) -[distributes]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=11230
(COMP) -[negatively_impacts]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=9915
(COMP) -[impact]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=9915
(PERSON) -[face]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=8984
(FIN_METRIC) -[impact]-> (ORG) -[discloses]-> (FIN_METRIC) | freq=7932
(PERSON) -[has_stake_in]-> (ORG) -[increase]-> (FIN_METRIC) | freq=6447
(PERSON) -[discloses]-> (ORG) -[increase]-> (FIN_METRIC) | freq=6134
```

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```
Base: (PERSON) -[has_stake_in]-> (ORG) -[discloses]-> (FIN_METRIC)
+ -[positively_impacts]-> (PRODUCT) | freq=120
+ -[impacted_by]-> (MACRO_CONDITION) | freq=72
+ -[increase]-> (ECON_IND) | freq=36
+ -[related_to]-> (FIN_INST) | freq=36
+ -[include]-> (FIN_INST) | freq=36
```

```
Base: (PERSON) -[discloses]-> (ORG) -[discloses]-> (FIN_METRIC)
+ -[impacted_by]-> (MACRO_CONDITION) | freq=78
+ -[positively_impacts]-> (PRODUCT) | freq=73
+ -[impacted_by]-> (RISK_FACTOR) | freq=39
+ -[impact]-> (CONCEPT) | freq=31
+ -[related_to]-> (PRODUCT) | freq=31
```

```
Base: (PERSON) -[announces]-> (ORG) -[discloses]-> (FIN_METRIC)
+ -[impacted_by]-> (MACRO_CONDITION) | freq=84
+ -[impacted_by]-> (RISK_FACTOR) | freq=42
+ -[impact]-> (CONCEPT) | freq=29
+ -[related_to]-> (PRODUCT) | freq=29
```

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BASELINE RAG

which regulatory bodies are named in connection with apple's disclosed net income, and what litigation prompted this?

Apple's net income is not directly regulated by any specific regulatory body as it's a financial metric that companies disclose voluntarily. However, Apple, like many multinational corporations, is subject to various regulations and oversight from several regulatory bodies in different countries related to their business operations, tax practices, data privacy, and consumer protection.

The litigation that often comes up in connection with Apple's net income primarily revolves around tax-related issues. For instance:

1. European Commission: In 2016, the European Commission ordered Ireland to recover up to €13 billion (\$15 billion) in back taxes from Apple due to state aid rules being broken by granting undue tax benefits to Apple for over two decades.
2. United States: The Senate Permanent Subcommittee on Investigations (PSI) launched an investigation into Apple's international tax strategies in 2013, which led to the release of the "Senate Report on Apple Inc." that criticized Apple for transfer pricing practices used to avoid paying billions in U.S. taxes.
3. India: In 2020, the Indian Tax Department imposed a tax demand of around \$650 million on Apple after a reassessment of its past profits, alleging that a significant portion of Apple's India-related income was shifted to lower-tax jurisdictions.
4. Other countries like Australia, France, and the UK have also been involved in tax disputes with Apple over the years.



EMBEDDING RAG

Q1: Which regulatory bodies are named in connection with Apple's disclosed net income, and what litigation prompted this?

Top 8 retrieved chunks (similarity scores: [0.69 0.686 0.685 0.683 0.68 0.678 0.677 0.677]):

- [1] Table of Contents ## UNITED STATES SECURITIES AND EXCHANGE COMMISSION Washington, D.C. 20549 ## Form 10-K (Mark One) x ANNUAL REPORT PURSUANT TO SECTI...
- [2] ## Table of Contents ## Apple eBooks Antitrust Litigation (United States of America v. Apple Inc., et al.) On April 11, 2012, the U.S. Department of J...
- [3] ## PART III ## Item 10. Directors, E xecutive Officers and Corporate G overnance The information required by this Item is set forth under the headings...
- [4] ## Available Information The Company's Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and amendments to rep...
- [5] ## SIG NATURES Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange A ct of 1934, the Registrant has duly caused this report...
- [6] ## SIG NATURES Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange A ct of 1934, the Registrant has duly caused this report...
- [7] ## Table of Contents ## SIG NATURES Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange A ct of 1934, the registrant has du...
- [8] ## Available Information The Company's Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and amendments to rep...

Tier 1 Answer (vector RAG):

The context does not contain the answer to the question about Apple's disclosed net income. It mentions various regulatory bodies such as the U.S. Securities and Exchange Commission (SEC) and the U.S. Department of Justice, but it does not provide information about Apple's net income. The litigation mentioned is the Appl e eBooks Antitrust Litigation, but it is not connected to disclosed net income in the provided context.

Q2: What financial market conditions does Apple disclose as negatively impacting its financial metrics?

Top 8 retrieved chunks (similarity scores: [0.79 0.789 0.765 0.747 0.744 0.742 0.741 0.739]):

- [1] ## Available Information The Company's Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and amendments to rep...
- [2] ## Available Information The Company's Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and amendments to rep...
- [3] ## Note 1 - Summary of Significant Accounting Policies Apple Inc. and its wholly-owned subsidiaries (collectively 'Apple' or the 'Company') designs, m...
- [4] ## Apple Inc. ## Notes to Consolidated Financial Statements ## Note 1 - Summary of Significant Accounting Policies ## Basis of Presentation and Prepar...
- [5] ## Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations This section and other parts of this Annual Report on...
- [6] ## Apple Inc. ## Notes to Consolidated F inancial Statements ## Note 1 - Summary of Significant Accounting Policies Apple Inc. and its wholly-owned su...
- [7] ## Backlog In the Company's experience, the actual amount of product backlog at any particular time is not a meaningful indication of its future busin...
- [8] ## Apple Inc. ## Notes to Consolidated Financial Statements ## Note 1 - Summary of Significant Accounting Policies ## Basis of Presentation and Prepar...

Tier 1 Answer (vector RAG):

The context does not contain the answer. It discusses Apple's products, services, and financial reporting, but does not mention specific financial market conditi ons that negatively impact its financial metrics.



GRAPH- RAG

Q: Which regulatory bodies are named in connection with Apple's disclosed net income, and what litigation prompted this?

Graph facts found (10):

- [ORG_REG] u.s. department of justice --regulates--> AAPL
- [ORG_REG] u.s. district court for the southern district of new york --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [ORG_REG] sec --regulates--> AAPL
- [GPE] ireland --regulates--> AAPL

Tier 2 Answer (graph-enriched RAG):

The regulatory bodies mentioned in connection with Apple are:

1. U.S. Department of Justice
2. U.S. District Court for the Southern District of New York
3. SEC (U.S. Securities and Exchange Commission)

The litigation that prompted this is the Apple eBooks Antitrust Litigation (United States of America v. Apple Inc., et al.), which was filed by the U.S. Department of Justice on April 11, 2012, alleging an unreasonable restraint of interstate trade and commerce in violation of §1 of the Sherman Act.

Note: The context does not explicitly mention Apple's disclosed net income. It provides information about the regulatory bodies and the litigation, but it does not connect this information to Apple's net income.

Q: What financial market conditions does Apple disclose as negatively impacting its financial metrics?

Graph facts found (10):

- [FIN_METRIC] production ramp --impacted_by--> AAPL
- [COMP] channel partner --negatively_impacts--> AAPL
- [CONCEPT] price reduction --negatively_impacts--> AAPL
- [CONCEPT] price reduction --negatively_impacts--> AAPL
- [FIN_METRIC] stock price --impacted_by--> AAPL
- [COMP] third-party developer --negatively_impacts--> AAPL
- [COMP] third-party developer --negatively_impacts--> AAPL
- [COMP] derivative counterparties --negatively_impacts--> AAPL

Tier 2 Answer (graph-enriched RAG):

Based on the GRAPH FACTS, Apple discloses the following financial market conditions as negatively impacting its financial metrics:

1. Price reduction: Apple's financial metrics are negatively impacted by price reductions, as indicated by the relationship [CONCEPT] price reduction --negatively_impacts--> AAPL.

Additionally, other factors that may negatively impact Apple's financial metrics include:

1. Production ramp issues, as indicated by the relationship [FIN_METRIC] production ramp --impacted_by--> AAPL.
2. Issues with channel partners, as indicated by the relationship [COMP] channel partner --negatively_impacts--> AAPL.

These relationships suggest that Apple's financial performance can be affected by various market and operational factors. However, the question specifically asks about financial market conditions, and price reduction is the most directly relevant factor disclosed.



GRAPH- RAG

```
=====
Q: Which regulatory bodies are named in connection with Apple's disclosed net income, and what litigation prompted this?

--- Tool calls made by agent ---
  Called: graph_traversal_tool | Input: {'input_str': 'AAPL|regulates'}
  Called: graph_traversal_tool | Input: {'input_str': 'AAPL|subject_to'}
  -> Result: [ORG_REG] u.s. department of justice --regulates--> AAPL
[ORG_REG] u.s. district court for the southern district of new york --regulates--> AAPL
[ORG_REG] sec --regulates--> AAPL
[ORG_REG] sec --regul
  -> Result: [ORG] board of director --subject_to--> AAPL
[ACCOUNTING_POLICY] asu 2016-02 --subject_to--> AAPL
[ORG] aapl --subject_to--> AAPL
[ORG] aapl --subject_to--> AAPL
[ORG] aapl --subject_to--> AAPL
[ORG]

Final Answer: The regulatory bodies named in connection with Apple's disclosed net income are the U.S. Department of Justice, the U.S. District Court for the South
ern District of New York, and the SEC. The litigation that prompted this disclosure is not specified in the provided information.
```

```
=====
Q: What financial market conditions does Apple disclose as negatively impacting its financial metrics?

--- Tool calls made by agent ---
  Called: graph_traversal_tool | Input: {'input_str': 'AAPL|negatively_impacts'}
  -> Result: [COMP] channel partner --negatively_impacts--> AAPL
[CONCEPT] price reduction --negatively_impacts--> AAPL
[CONCEPT] price reduction --negatively_impacts--> AAPL
[FIN_MARKET] liquidity --negatively_im
  Called: graph_traversal_tool | Input: {'input_str': 'AAPL|impacted_by'}
  -> Result: [FIN_METRIC] production ramp --impacted_by--> AAPL
[FIN_METRIC] stock price --impacted_by--> AAPL
[MACRO_CONDITION] consumer demand --impacted_by--> AAPL
[LOGISTICS] supply chain --impacted_by--> AAPL

Final Answer: Apple discloses that the following financial market conditions negatively impact its financial metrics:
- Price reduction
- Liquidity
- Consumer demand
- Supply chain
- Sale and distribution channel
- Service offering
- Foreign currency weakness
PS A:\S7\Phase Project> █
```

